

Week 3 Report: Financial Data Analysis using Power BI (Shopify Stock Data)

Arni G
BSC DS &AI

1. Introduction

In this week's task, financial data analysis was performed on Shopify stock data using Power BI. The objective was to analyze stock performance, trading volume, and returns using DAX measures and interactive visualizations.

2. Data Import and Transformation

The dataset was imported into Power BI using the CSV file option. In Power Query, the following transformations were applied:

- Converted the date column to **Date format**
- Set open, high, low, and close columns to **Decimal Number**
- Converted volume to **Whole Number**
- Removed unnecessary columns and cleaned the dataset

date	open	high	low	close	adj_close	volume
21 May 2015	2.79999995231628	2.87400007247925	2.41100001335144	2.56800007820129	2.56800007820129	123039000
22 May 2015	2.60700011253357	3.10999989509583	2.59999990463257	2.83100008964539	2.83100008964539	28412000
26 May 2015	2.98000001907349	3.03399991989136	2.90799999237061	2.96499991416931	2.96499991416931	8202000
27 May 2015	3.06699991226196	3.08100008964539	2.70000004768372	2.75	2.75	7976000
28 May 2015	2.75500011444092	2.7739999294281	2.64800000190735	2.74499988555908	2.74499988555908	4000000
29 May 2015	2.72499990463257	2.73799991607666	2.68700003623962	2.72000002861023	2.72000002861023	2554000
01 June 2015	2.72000002861023	2.74399995803833	2.66000008583069	2.73300004005432	2.73300004005432	2569000
02 June 2015	2.74699997901917	2.74699997901917	2.66499996185303	2.6949999332428	2.6949999332428	3798000
03 June 2015	2.67600011825562	2.75	2.65000009536743	2.74399995803833	2.74399995803833	3024000
04 June 2015	2.74200010299683	2.74900007247925	2.55100011825562	2.67799997329712	2.67799997329712	3226000
05 June 2015	2.66599988937378	2.73900008201599	2.60999989509583	2.73900008201599	2.73900008201599	1294000
08 June 2015	2.69000005722046	2.74799990653992	2.68199992179871	2.69799995422363	2.69799995422363	2809000
09 June 2015	2.66400003433228	2.84599995613098	2.66400003433228	2.83400011062622	2.83400011062622	3375000
10 June 2015	2.84500002861023	2.88800001144409	2.79500007629395	2.80999994277954	2.80999994277954	2734000
11 June 2015	2.84899997711182	3.07299995422363	2.84899997711182	3.07299995422363	3.07299995422363	4606000
12 June 2015	3.13199996948242	3.36999988555908	3.125	3.36299991607666	3.36299991607666	8474000
15 June 2015	3.3970000743866	3.81200003623962	3.29200005531311	3.54999995231628	3.54999995231628	16943000
16 June 2015	3.60599994659424	3.83789992332458	3.58400011062622	3.81100010871887	3.81100010871887	11095000
17 June 2015	3.96300005912781	4.21299982070923	3.80100011825562	3.86299991607666	3.86299991607666	15952000
18 June 2015	3.84200000762939	3.88899993896484	3.4326000213623	3.5550000667572	3.5550000667572	11482000
19 June 2015	3.63299989700317	3.67400002479553	3.25500011444092	3.55200004577637	3.55200004577637	9797000
22 June 2015	3.67600011825562	3.84999990463257	3.60899996757507	3.69400000572205	3.69400000572205	7922000
23 June 2015	3.77500009536743	3.80999994277954	3.51699995994568	3.54299998283386	3.54299998283386	2939000
24 June 2015	3.51200008392334	3.74289989471436	3.41400003433228	3.59599995613098	3.59599995613098	5644000

3. Date Table Creation

A Date Table was created using DAX to enable time-based analysis:

Dim_Date = CALENDARAUTO()

Additional columns such as Year, Quarter, Month, and Day were created. The Month column was sorted using Month Number, and the table was marked as a Date Table.

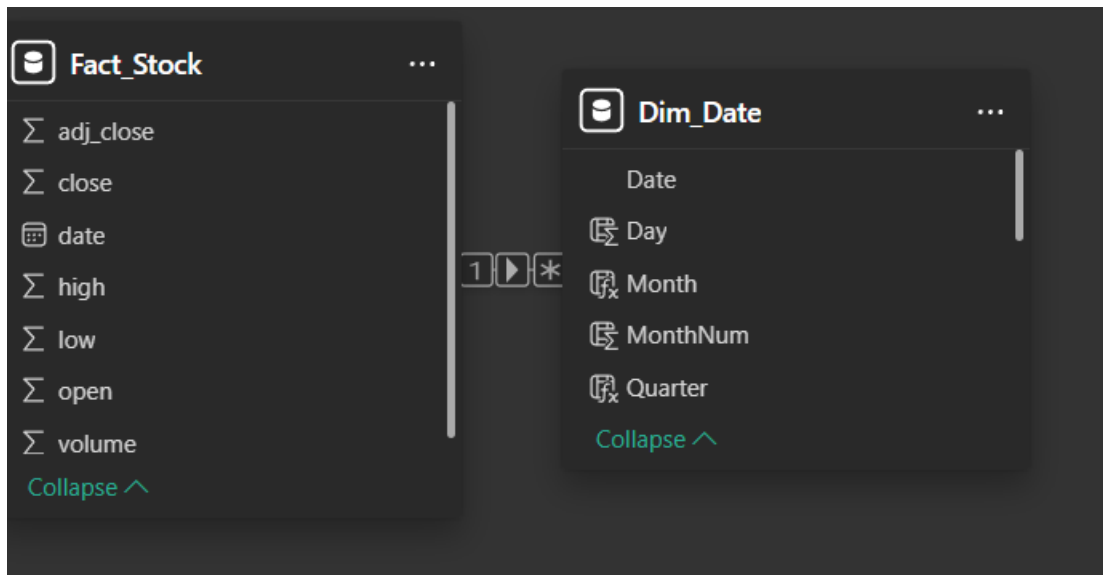
1 Dim_Date = CALENDARAUTO()						
Date	Year	Quarter	MonthNum	Month	Day	
01-01-2015 00:00:00	2015	Q1	1	Jan	1	
02-01-2015 00:00:00	2015	Q1	1	Jan	2	
03-01-2015 00:00:00	2015	Q1	1	Jan	3	
04-01-2015 00:00:00	2015	Q1	1	Jan	4	
05-01-2015 00:00:00	2015	Q1	1	Jan	5	
06-01-2015 00:00:00	2015	Q1	1	Jan	6	
07-01-2015 00:00:00	2015	Q1	1	Jan	7	
08-01-2015 00:00:00	2015	Q1	1	Jan	8	
09-01-2015 00:00:00	2015	Q1	1	Jan	9	
10-01-2015 00:00:00	2015	Q1	1	Jan	10	
11-01-2015 00:00:00	2015	Q1	1	Jan	11	
12-01-2015 00:00:00	2015	Q1	1	Jan	12	
13-01-2015 00:00:00	2015	Q1	1	Jan	13	
14-01-2015 00:00:00	2015	Q1	1	Jan	14	
15-01-2015 00:00:00	2015	Q1	1	Jan	15	
16-01-2015 00:00:00	2015	Q1	1	Jan	16	
17-01-2015 00:00:00	2015	Q1	1	Jan	17	
18-01-2015 00:00:00	2015	Q1	1	Jan	18	
19-01-2015 00:00:00	2015	Q1	1	Jan	19	
20-01-2015 00:00:00	2015	Q1	1	Jan	20	
21-01-2015 00:00:00	2015	Q1	1	Jan	21	
22-01-2015 00:00:00	2015	Q1	1	Jan	22	
23-01-2015 00:00:00	2015	Q1	1	Jan	23	
24-01-2015 00:00:00	2015	Q1	1	Jan	24	
25-01-2015 00:00:00	2015	Q1	1	Jan	25	
26-01-2015 00:00:00	2015	Q1	1	Jan	26	
27-01-2015 00:00:00	2015	Q1	1	Jan	27	

4. Data Modeling

A relationship was established between:

- Dim_Date[Date] → Fact_Stock[date]

This enabled proper filtering and supported time intelligence calculations.

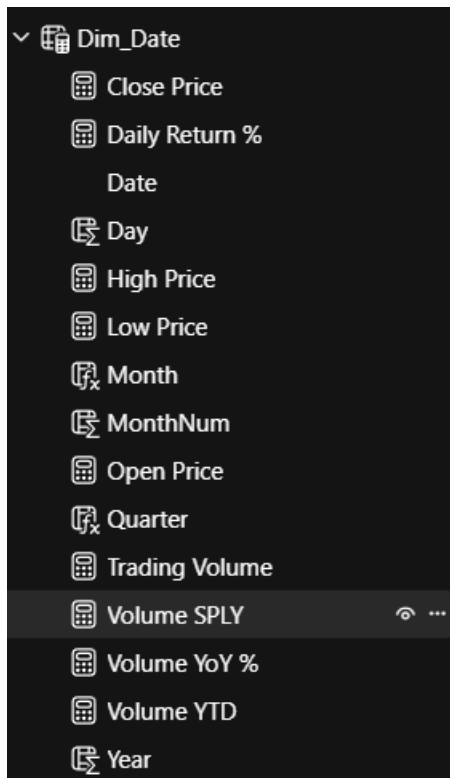


5. DAX Measures

The following measures were created:

- **Open Price**
- **Close Price**
- **High Price**
- **Low Price**
- **Trading Volume**
- **Daily Return %**
- **Volume YTD (Year-to-Date)**
- **Volume YoY % (Year-over-Year)**

These measures helped analyze stock performance effectively.



6. Data Visualization (Based on Dashboard)

The dashboard titled “**Shopify Stock Market Analysis Dashboard**” includes:

Card Visuals:

Daily Return % = **0.87**

Volume YTD = **41bn**

Pie Chart:

Shows comparison of **High, Low, and Open Prices**

Treemap:

Displays **Open Price and Low Price by Quarter**

Bar Chart:

Shows **Trading Volume by Month**

Line Chart:

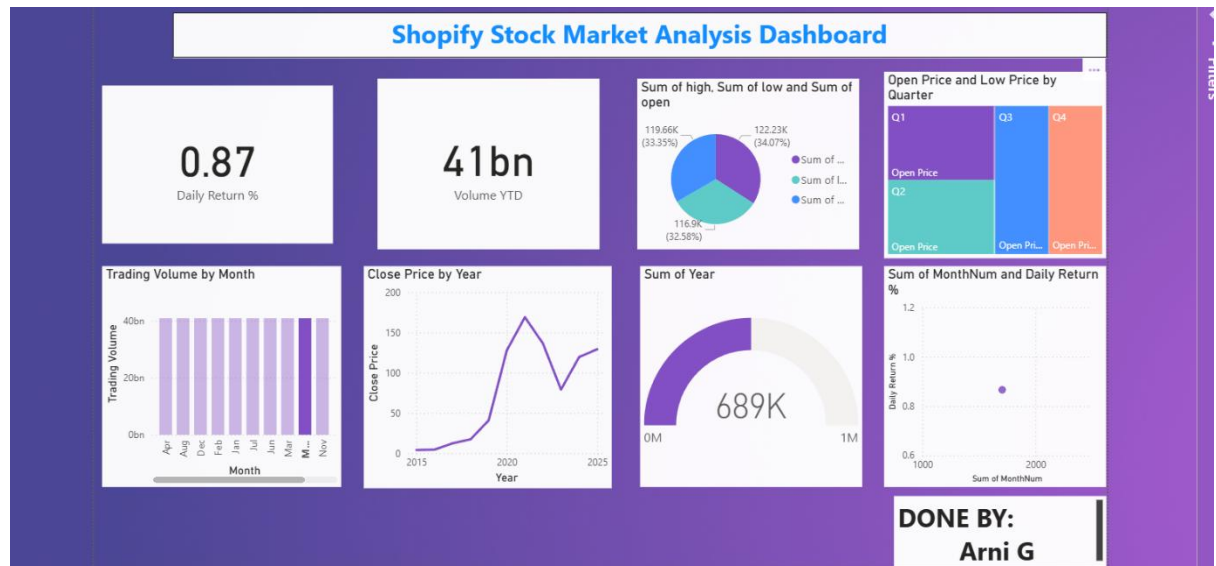
Displays **Close Price trend over years**

Gauge Chart:

Shows total yearly value (**689K**)

Scatter Plot:

Shows relationship between **Monthly data** and **Daily Return %**



7. Insights

- The stock price shows fluctuations over the years with noticeable peaks.
- Trading volume remains relatively consistent across months.
- Daily return percentage indicates moderate performance.
- Quarterly analysis helps identify variation in open and low prices.

8. Conclusion

The Power BI dashboard successfully provides insights into Shopify stock performance. Using DAX measures and time intelligence functions, the analysis helps understand trends, volume patterns, and financial performance effectively.